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Research Question:

To what extent has Argentina suffered more from the effects of inflation compared to the developed G7 countries, during the period 2018-23?

Introduction

The global rise in inflation surrounding the Covid-19 period (2018-2023) disproportionately affected nations with different development levels. We will focus on the G7, showing a good sum of developed countries, (Canada, France, Germany, Italy, Japan, UK and US). These will allow us to compare to developing nation of Argentina across several macroeconomic indicators. Argentina is a country with high inflation that's persistent and uncontrollable, subsequently damaging several economic factors. Whereas developed nations can curb inflation and recover. This project aims to show how and why it is true that poorer countries suffer unequally from the effects of rising inflation, and the overwhelming effect of the pandemic on the G7 and Argentina's macroeconomic progress.

Literature

1. EU Science Hub (2022): https://joint-research-centre.ec.europa.eu/jrc-news-and-updates/inflation-increases-poverty-unevenly-widening-gaps-across-eu-2022-12-06_en#:~:text=As%20current%20inflation%20is%20driven,spending%20on%20such%20essential%20items

This article addresses our question, highlighting how inflation increases poverty and widens the gap between developed and developing countries, within the EU. It shows the impact on poorer households, who allocate a larger portion of their income to essential commodities like food due to rising prices. The commodity prices will be a topic we focus on.

2. World economic forum (2022): <https://www.weforum.org/agenda/2022/02/explainer-how-is-inflation-hitting-low-income-households-in-developing-nations/>

In Argentina, inflation has increased to over 100%, triggering economic crises. There is a major loss in confidence in the peso, leading to significant depreciation and large central bank debt. We will mention the

unsuccessful and damaging policy that has left both consumers and businesses grappling with the challenges of volatile and unpredictable pricing.

3. National library of medicine (2022) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8899350/>

This article shows clear global inequality, with 69% of respondents in poorer countries experiencing income drops compared to 45% in richer ones. It highlights how trade and tourism disruptions worsened recovery and deepened inequalities in poorer nations due to the closing of sectors, highlighting the harsh impact of Covid-19.

Our research uncovered various theories regarding the effects, responses, and recovery attempts from Covid-19. However, we noticed a gap in the literature regarding summary of the macroeconomic effects. We aim to use data to support these claims and demonstrate that poorer countries like Argentina suffered uncontrollably from the fallout of the Covid-19 inflationary period, not only due to their economic status, but their responses.

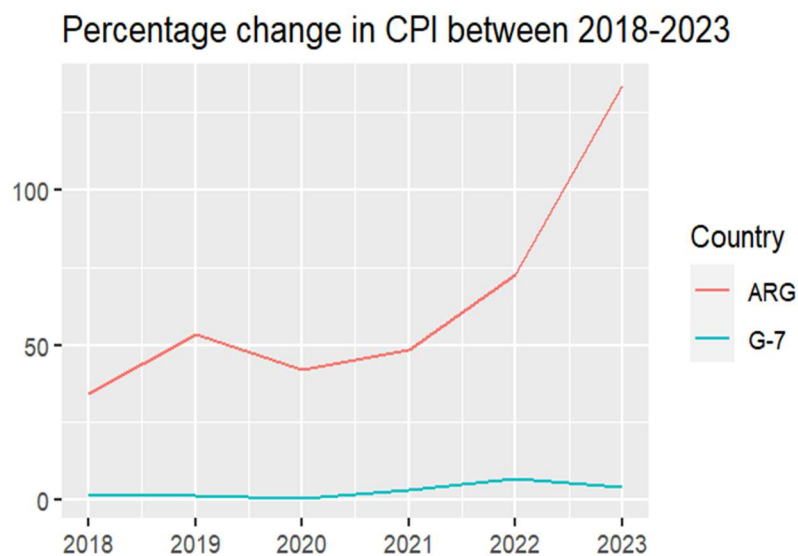
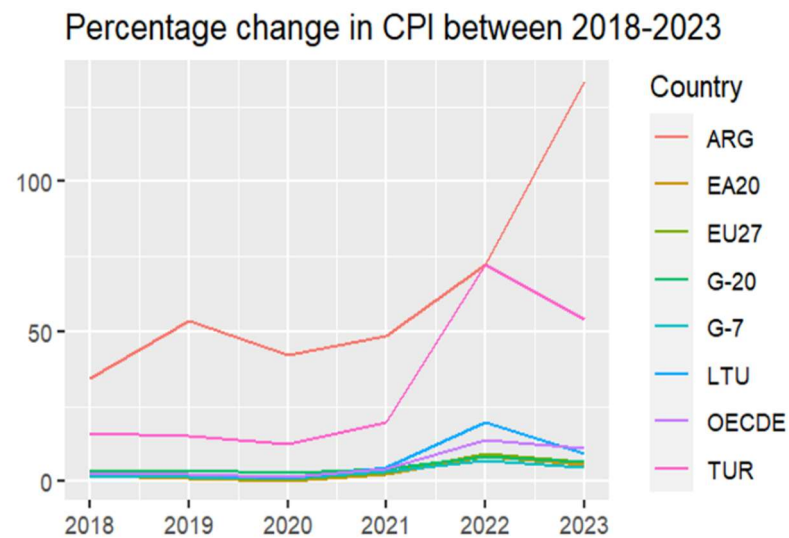
Data and Methods

Starting in 2018, leading into the Covid-19 crisis of 2019-2020 and ending in 2022/23, we will analyse several variables. At first, we visualised the inflation rates over the period, this has been achieved by using the annual percentage change of CPI across countries. By primarily showing the changes in inflation over the period, this helps to illustrate how, as every country suffered from Covid-19, some suffered more than others in terms of larger and more volatile inflation. This initial exploration into the growth of inflation rates will allow to correlate between other economic indicators. Overall, we are focussing on showing inflation rate, poverty, economic growth and unemployment. Subsequently mentioning, how and why Argentina failed to control the economic challenges.

A key macro-economic variable that will be analysed is GDP growth, both annually and quarterly to see a more detailed breakdown over the 5-years surrounding the Covid-19 pandemic. Once again, to keep a fair and structured comparison, by using the Gross Domestic Product, summarising a countries output, we can compare the extent of the fall and rise in production and economic output. As it is clear some countries are much larger and produce much more in terms of revenue and volume, we have chosen to focus on the percentage change in GDP per period, therefore it is relative and shows the comparative national struggles to analyse and compare this. Essentially, we are aiming to compare economic growth mainly during, and after the pandemic to help answer our question and show if different levels of developed countries respond and react from economic adversity at much differing rates. The analysis of this data is crucial in terms of key macroeconomic indicators of development, and by choosing to use percentages as it will highlight economic growth and relative changes.

Results

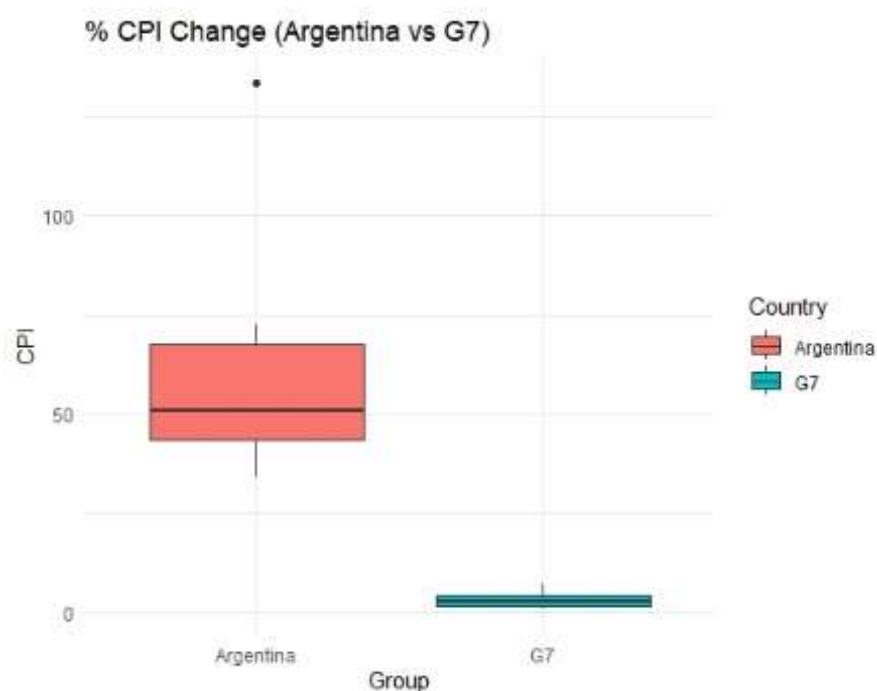
I) Inflation



Using OECD data, it is clear Argentina's CPI change year-on-year is the most volatile. In periods such as 2022, where all areas suffered a rise in inflation, the relative change in developing countries shown above, is much greater, reaching up to 133%. Whereas the blocks such as G20 and OECD managed to control inflation and remain stable over the period

	Country	2018	2019	2020	2021	2022	2023	region
1	ARG	34.2772300	53.5483100	42.01509000	48.4093800	72.430760	133.4889000	Argentina

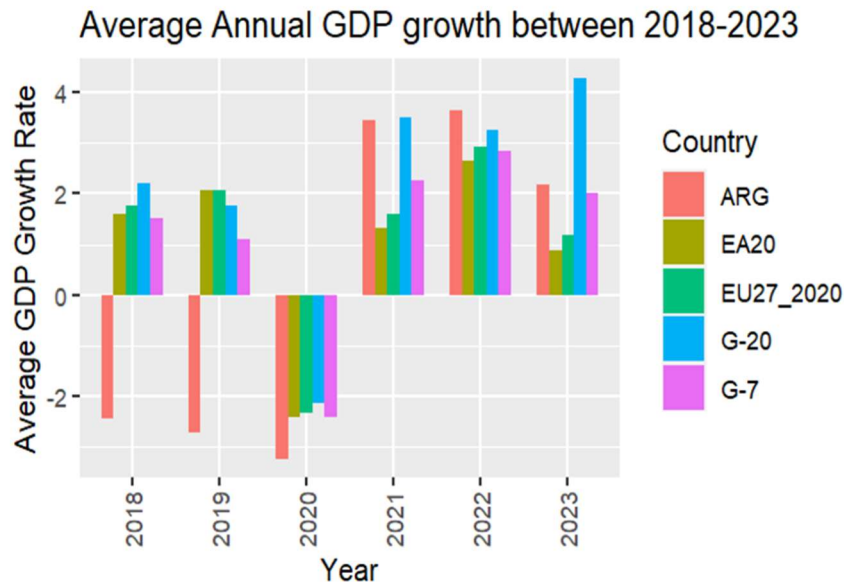
This reinforces the point that developing economies suffer more as inflation is higher long term and they're unable to reduce this immediately. In Argentina, where inflation continues to steeply rise, this has been attributed to many factors such as monetary policy. A reason why Argentina suffered compared to G7, was their response. Looking at the second graph, from 2018-19 then 2020 onwards, inflation in Argentina rapidly increased, whereas G7 never had any issues. In 2018 the IMF granted a \$57 billion relief fund to help clear national debt in the crisis, however this was unsuccessful and carried over. During Covid-19, due to limited finances and missed payments to creditors, the government decided to print money (QE) to help their citizens. This implementation severely increased inflation through the flow of money in the economy. Furthermore, a drought caused supply shortages in crops, hence a further rise in food prices. Due to these factors, Argentina have been constantly fighting to hike interest rates to curtail inflation.



It is clear when analysing box plots, that not only is the high level of inflation a problem, but also the variability. Argentina faces an average CPI change of 64.03% compared to the G7 average of 3.26. Furthermore, the standard deviations are 36.4 and 2.3 respectively. Due to developed economies being able to mitigate inflation through responses such as fiscal policy, their inflation rates are much lower, as are their deviations and long-term volatility.

II) GDP

As Covid-19 became a global crisis in 2020, most countries experienced similar economic cycles. The data highlights Argentina's economic volatility: it faced the largest GDP decline in 2020 but then showed the highest growth rates during the recovery period from 2021 onwards.

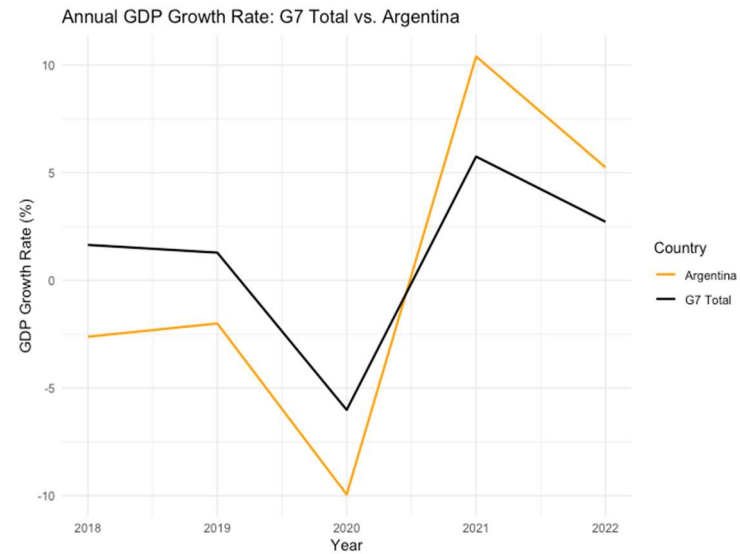
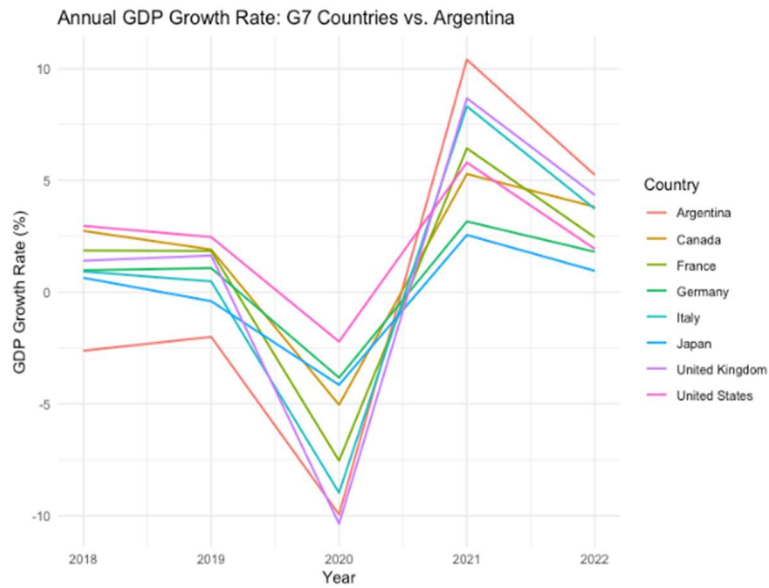


Argentina Compared to G7 Countries (Annually):

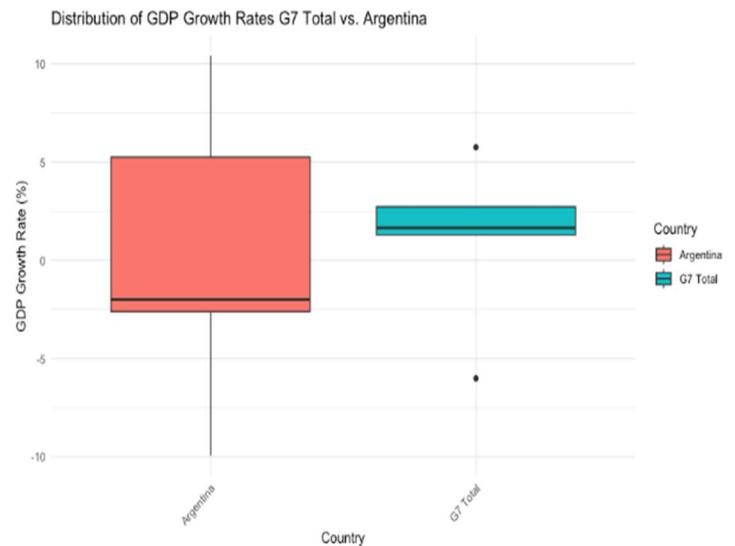
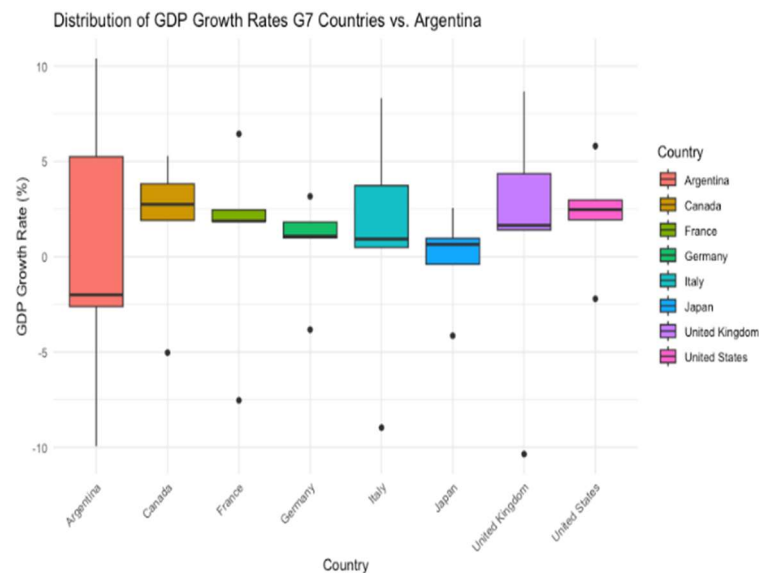
Pre-pandemic, Argentina faced economic challenges, reflected in a contractionary phase with average GDP growth rates of -2.4% in 2018 and -2.75% in 2019. In stark contrast, the G7 nations, maintained positive growth rates, averaging between 1.2% to 2.2% during the same period, indicating a more stable pre-pandemic economic landscape.

The pandemic made economies worldwide grind to an abrupt halt, leading to significant GDP declines. Argentina registered one of the largest GDP contractions, underscoring the severity of its economic volatility.

In 2021, Argentina recorded positive GDP growth following three consecutive years of decline, signalling a potential turnaround with a 10% growth. This coincided with global easing of pandemic restrictions, (reopening of commerce and hospitality sectors) and a broader economic recovery, with the G7 nations averaging a 5.7% growth.



When observing the combined mean growth rate of the G7 nations, a pattern of less pronounced fluctuations becomes evident. Despite facing the same cyclical downturn due to the pandemic, the G7's recovery trajectory was marked by stability, showing their robust economic structures capable of withstanding external shocks. Using a box plot to visually represent this. It is evident that all the G7 nations during the period experienced a positive average growth.



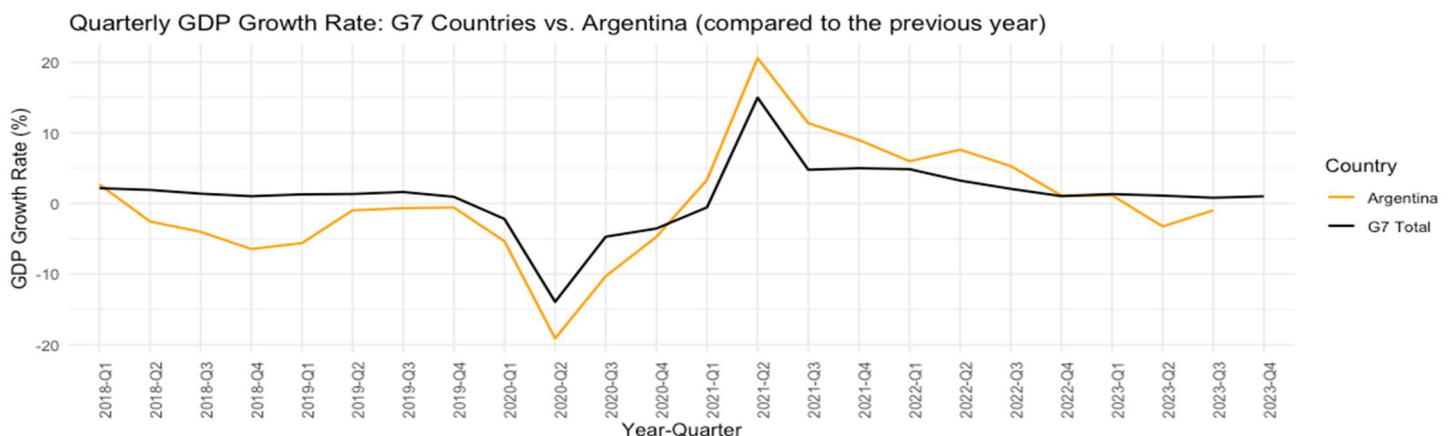
Analysing comparative statistics. Argentina has a higher standard deviation compared to the G7 total, implying Argentina's economic growth rate is more volatile. High volatility is a sign of an economy that is more susceptible to external shocks and internal economic issues, both of which hampered Argentina.

	Country	Mean	Median	SD	Min	Max
1	Argentina	0.2159604	-2.000861	7.827867	-9.943235	10.39825
2	G7 Total	1.0777579	1.646966	4.335098	-6.014306	5.74774

Argentina compared to G7 Countries Quarterly:

When examining GDP growth on a quarterly basis, the underlying economic trends, and responses to external shocks such can become clearer.

For Argentina, a quarterly analysis shows more immediate impacts of government interventions. Given Argentina's high volatility annually, its quarterly figures display sharper declines and more rapid recoveries, mirroring the intense economic instability. The trends may be similar but Argentina always rises/falls more.

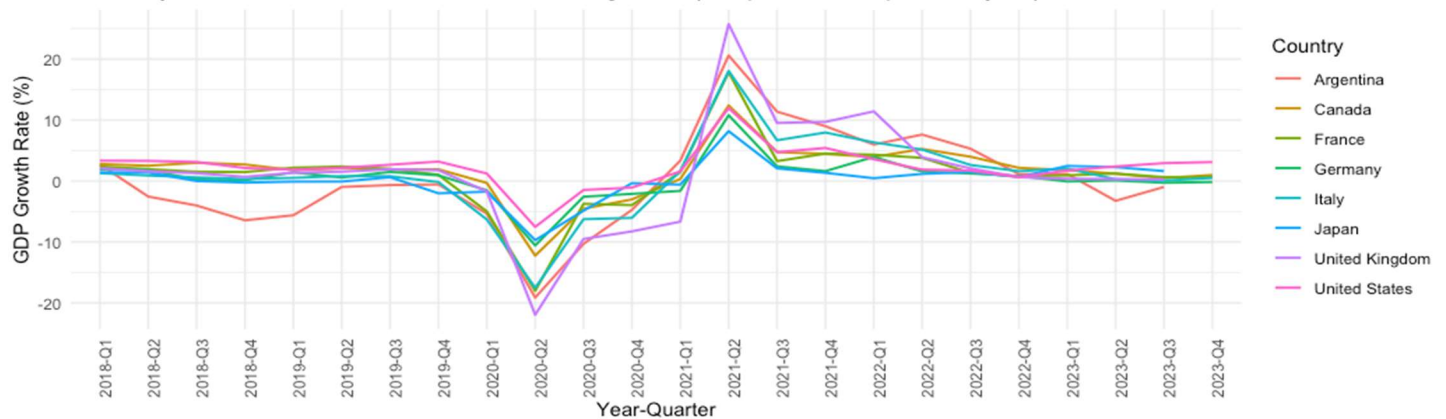


Conversely, G7 countries, with their diversified and resilient economies, show quarterly GDP figures that are less erratic. While still reflecting the global economic downturn and subsequent recovery phases, the fluctuations in the G7's quarterly GDP growth rates are less extreme. This stability can be attributed to stronger institutional frameworks, more substantial financial markets, and broader economic bases, which help buffer against the same shocks.

	Country	Mean	Median	SD	Min	Max
1	Argentina	0.1561042	-0.6658787	7.999922	-19.12028	20.58639
2	G7 Total	1.1241089	1.3069310	4.838534	-13.93253	14.98767

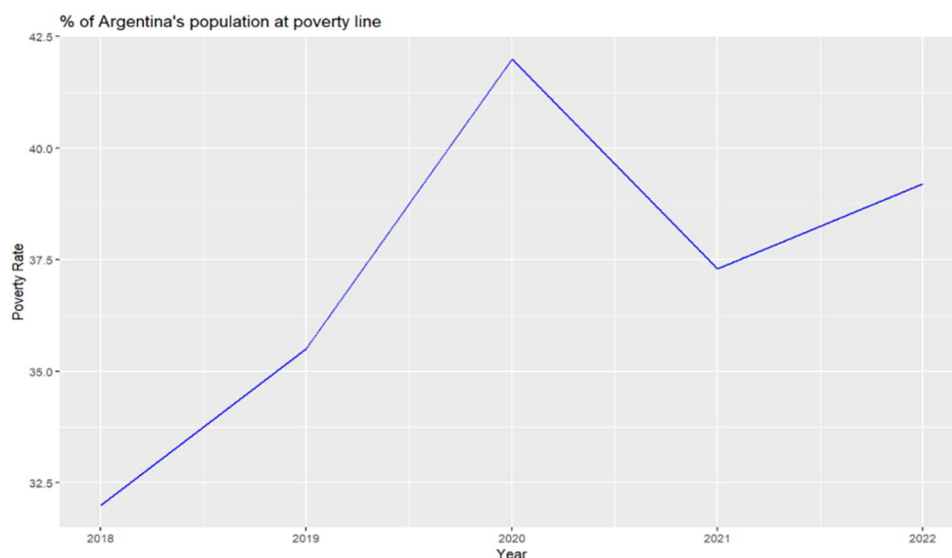
Argentina, during the 5-year period, experienced fluctuations ranging from -19.1% (2020-Q2) to 20.6% (2021-Q2) GDP growth while the G7 nations -13.9% and 15% respectively.

Quarterly GDP Growth Rate: G7 Countries vs. Argentina (compared to the previous year)

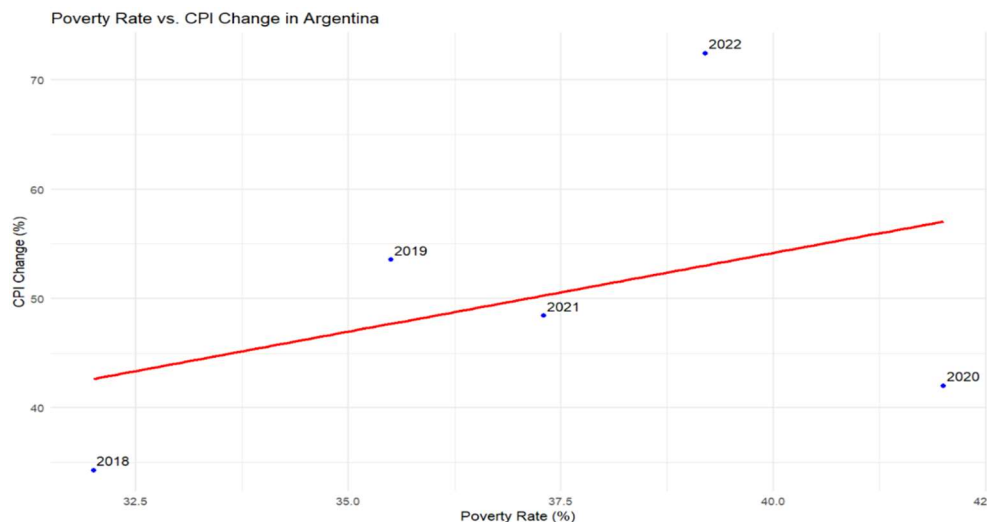


Argentina was significantly affected by the Covid-19 pandemic, particularly during the critical years of 2019-2020. While Argentina's GDP experienced a dramatic decline, it also rebounded sharply, in times of resurgence that can be attributed to the encouragement of spending post-lockdown. Whereas, G7 nations encountered lesser setbacks, returning to a stable growth level. This pattern mirrors trends seen in inflation, illustrating Argentina's ongoing struggles to stabilize its macroeconomic environment.

III) Poverty in Argentina



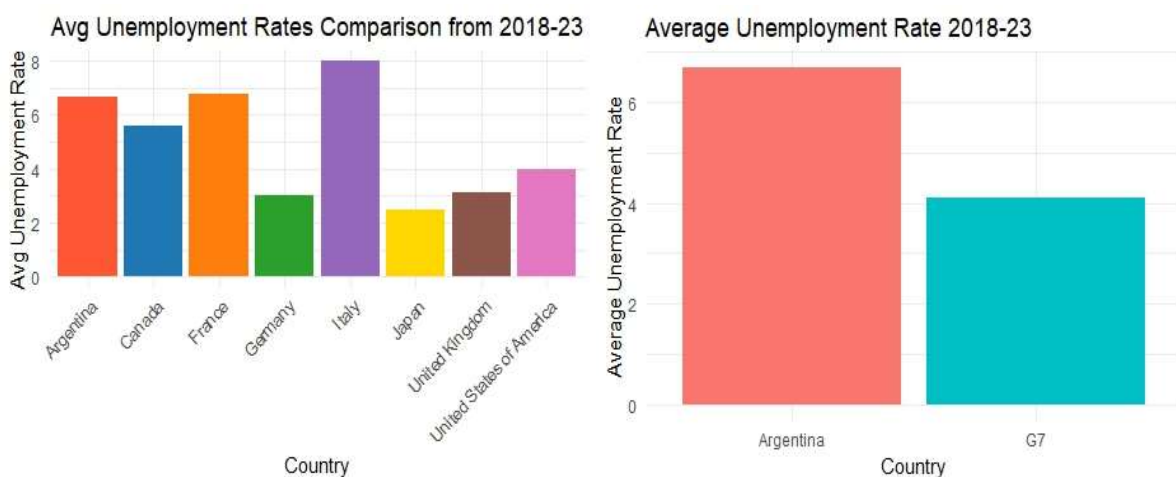
Resultingly from high inflation, Argentina suffered a large cost of living crisis with many falling into poverty. This is a snowball effect from changes in food prices (part of CPI) and average GDP, previously mentioned, creating uncertainty, and forcing daily predicaments of what to buy and where to find affordable goods.



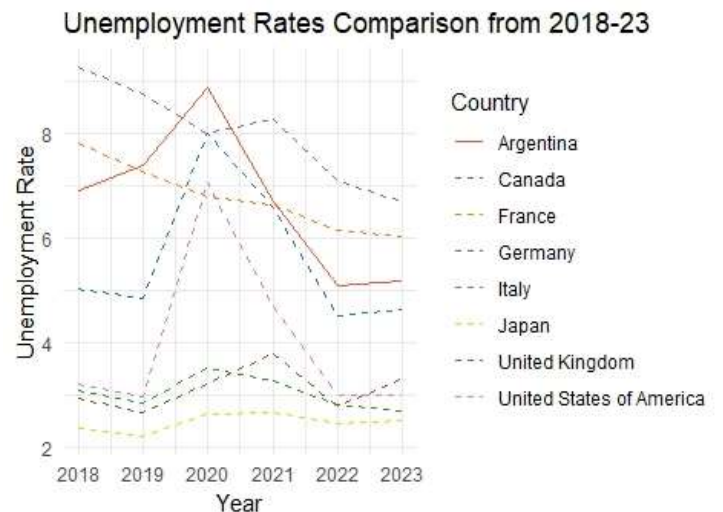
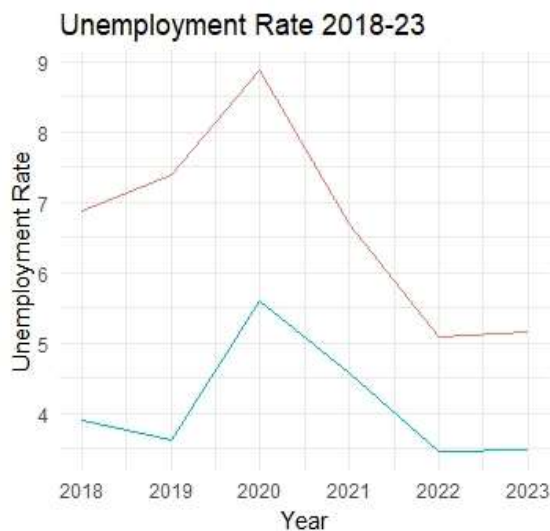
There is a visual positive correlation between poverty and inflation, calculating the correlation coefficient obtained the value 0.3778, reflecting a weak positive correlation, perhaps less than hypothesised, but can be in-part attributed to a small data set. The large rise in proportion at the poverty line from 32-42% (2018-2020), then reaching a 39.2% in 2022, shows a clear increasing socio-economic issue.

A significant reason for the inflation and cost of living crisis is the government's response policy. The continued devaluation of the Peso has led to a sharp rise in import prices including, utility, oil, and food. The excessive stimulus to aggregate demand through quantitative easing, government expenditure and devaluations, have all led to soaring inflation rates and therefore, Argentinians are struggling to make ends meet. Salaries and pensions cannot keep up with the inflation trend and there have been calls to replace the Peso with the dollar to stabilise and avoid hyperinflation. The policy responses to attempt a decrease in inflation and recovery in Argentina are not working, the nation and its economy is severely suffering on a cost-of-living basis, highlighted in the period 2018-2023, with many unable to afford commodity food prices.

IV) Unemployment



Argentina has always seemed to be battling with high unemployment rates, we can see that Argentina's unemployment rate is most commonly at 7% which is almost double the 3.5% of the G7 countries.



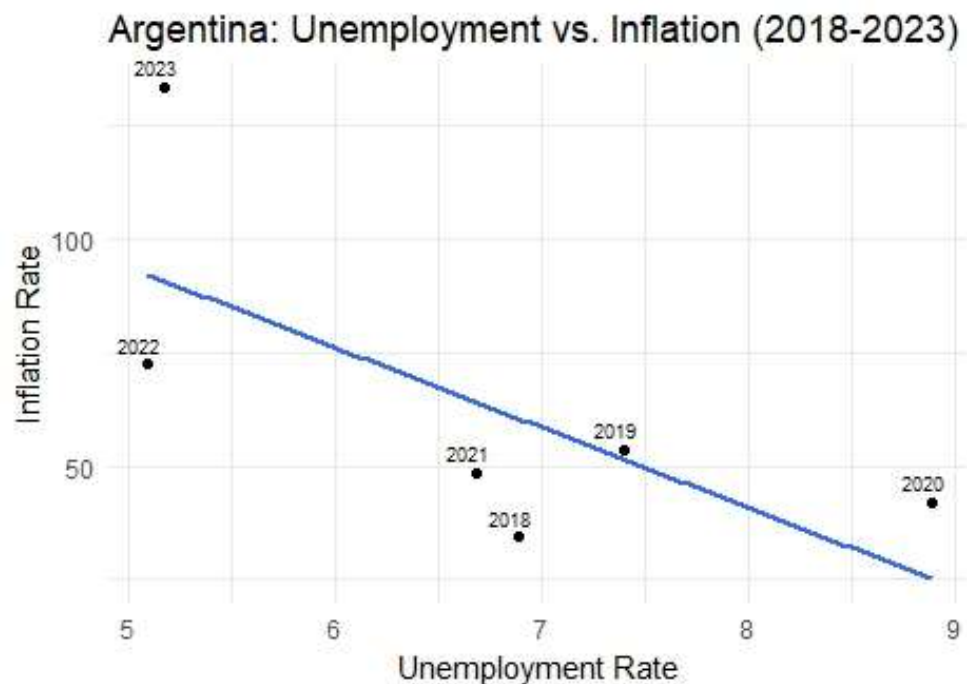
In the time series graph, we note that Argentina's unemployment rate was rising from 2018 onwards from 7% - 7.5%, between 2018-2019 which was pre-pandemic, whereas the G7 countries saw a reduction from about 4% - 3.5% in the same period. Their rising unemployment rates pre pandemic could be due to their agreement with the IMF in 2018, where Argentina committed to reducing its fiscal deficit through austerity measures, which included cuts in public sector spending and subsidies. These cuts often resulted in public sector layoffs and reduced economic activity. Furthermore, the devaluation of the Argentinian Peso in 2018, led to a higher cost for business and lower investor confidence, once again increasing unemployment.

Although Argentina's unemployment rate peaked at the start of 2020, surpassing all the G7 countries, surprisingly, the Covid-19 pandemic had a bigger impact on G7 unemployment than in Argentina. From the time series graphs we observe that on average unemployment rose by around 1.5% in Argentina during 2019 but around 2% in the G7 countries. This can be attributed to G7 countries having larger service sectors. The service sector industries were more impacted by lockdown measures, such as the furlough scheme. Argentina has a more diversified economy, with large parts of their GDP coming from the agricultural sector, which was not as affected by lockdown regulations.

We can also observe a sharp reduction in Argentina's unemployment rate from 2020 - 2022, which was lower than pre Covid-19 times and had a much sharper decrease than G7 countries. As mentioned earlier, post pandemic certain sectors that were the main drivers of their economy such as agriculture and exports were able to rebound faster, thus people were quickly re-employed into these sectors. It is important to note that post-crisis labour markets often see adjustments where workers accept lower wages or reduced

hours, which can technically reduce unemployment numbers but may not fully represent economic recovery.

Phillips curve is an economic theory which shows inverse relationship between unemployment and inflation rates. We have created a linear regression model to understand the relationship between unemployment and inflation rates in Argentina. For our model, unemployment is taken as independent variable and inflation as dependent variable.



The correlation coefficient between Argentina's unemployment and inflation rates was calculated to be approximately -0.695. This indicates a moderate to strong negative linear relationship between them. The negative sign indicates that there is a relatively strong tendency for unemployment and inflation rates to move in opposite direction for Argentina. This aligns with the typical expectation described from the Phillips curve framework. However, this relation doesn't necessarily imply that changes in one variable will directly affect the other variable. Other factors and socio-economic policies may also influence the variables.

We can gain insights into the complex interplay between unemployment and inflation in Argentina by understanding the summary of this linear model. The intercept talks about the value of dependent variable (inflation rate) when all independent variables (unemployment rate) are zero. So, the intercept is 182.374 but it may not have a meaningful interpretation as unemployment rate is unlikely to be zero in the real world. On the other hand, the coefficient for the unemployment rate is -17.704. This shows, an increase in one-unit of unemployment rate causes on average a decrease of -17.704 units in the inflation rate. Overall, the intercept and coefficient help us understand how changes in unemployment may impact inflation, allowing policymakers and economists to make decisions and predictions about economic trends and policies.

Conclusion

Upon visualising macroeconomic variable data, it's clear Argentina's economy is still unstable exiting the COVID years, the reasons why can be attributed to their policy responses in recent years, stemming from political instability. We've seen how steep inflation translated into cost-of-living crisis' and falling economic growth, together with poor responses have left the country in turmoil. Conversely, the strong, responsive, and adaptable G7 countries have shown, using the data, to recover quickly from the pandemic's downturns and challenges, now surpassing even pre-pandemic levels. The models reflect overall global trends, however Argentina experiencing greater volatility in terms of higher highs and lower lows, suffering to a much larger extent.

Data sources

i) Inflation: OECD data- https://stats.oecd.org/Index.aspx?DataSetCode=PRICES_CPI

ii) GDP: OECD data- <https://stats.oecd.org/index.aspx?queryid=60703>

iii) Poverty: World bank data - <https://data.worldbank.org/indicator/SI.POV.NAHC?locations=AR>

iv) Unemployment-

https://rshiny.ilo.org/dataexplorer24/?lang=en&segment=indicator&id=UNE_2EAP_SEX_AGE_RT_A

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